What are the implications of computational complexity for law and economics?

Notes for Book Project

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Yale, Columbia and NBER

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Introduction

Introduction

- This lecture is based on preliminary work for a book project on “Law and Economics in the Age of Computers.”
- I look forward to your comments and suggestions.

Goals of Lecture I

- The law and economics movement has helped organize and improve the law across a wide variety of areas.
- For the most part, this has relied on price theory, but there are a number of questions that price theory cannot fully address.
- This lecture breaks down the analytical tools of economics into three distinct classes of models:
    - Price theory (*old school economics*)
    - Information economics (*new school economics*)
    - Bounded Rationality (*contemporary economics and computer science*)

Goals of Lecture II

- Each class of models provides a different lens through which to analyze legal and economic issues.
- The choice of model depends on the specific legal and economic issue being analyzed, and whether the framework can lead to better decisions.
- I aim to provide a formal model of bounded rationality consistent with H.L.A. Hart's account of the "penumbra of uncertainty" in legal rules.
- This provides a foundation for understanding the importance of values and their role in the design of legal rules that accommodate bounded rationality while remaining effective and practically enforceable.

The Origins of Law

- All complex human societies have developed legal systems to regulate behavior, resolve disputes, and maintain social order.
- These systems have evolved over time to address the changing needs and complexities of human societies and, in the sense of Posner (1973), can be viewed as improving social outcomes over time.

Hammurabi's Code



- Hammurabi's Code is one of the earliest legal codes, dating back to around 1750 BC in ancient Mesopotamia. It is a surprisingly comprehensive set of laws that covers various aspects of society, including family, property, immorality, theft, burglary, wage and price regulation, and even professional conduct.
- See Vincent (1904) for a comprehensive discussion of the code.

Implications of Hammurabi's Code for Today

- The code is famous for its principle of "an eye for an eye," which emphasizes the idea of retributive justice.
- It also includes provisions for social hierarchy, with different punishments based on the social status of the offender and the victim.
- What is interesting is that many of the main issues have not changed in almost 3,800 years!
- What has changed:
  - New technologies that require new notions of property rights.
  - Better economic theory and evidence to guide the design of better rules (the goal of this lecture).

Law and Economics: Old School and New School Economics

Old School Economics is not so old.

- In contrast to the law, economics is a very new discipline. What one might consider *modern economics* had its origins in the 1950s, with the development of general equilibrium theory providing a rigorous mathematical framework for price theory.
- The law and economics movement also began at that time with the founding of the *Journal of Law and Economics* by Aaron Director and Ronald Coase at the University of Chicago.
- The work of Posner (2014), Baird and Gertner (1994), Cooter and Ulen (2016), and Shavell (2004) has been influential in bringing price theory and information economics to law schools.
- See Zamir and Teichman (2018) integrating behavioral economics into law and economics.

For the purposes of this lecture, Economics is divided into three groups:

- Price Theory and Game Theory (1940-1980) - the foundations of modern economics (see the appendix of MacLeod (2022)).
- Information Economics, Contract Theory and Causal Analysis (1980-2010)
- Bounded Rationality, Behavioral Economics and Big Data (ideas dating to Simon (1955), but only computationally operational from roughly 2010 on)

New School Economics: Failures of Price Theory due to Imperfect Information i

Year	Laureates	Contribution Summary
2001	George A. Akerlof, A. Michael Spence, Joseph E. Stiglitz	For fundamental analyses of markets with asymmetric information (when buyers or sellers have unequal information).
2007	Leonid Hurwicz, Eric S. Maskin, Roger B. Myerson	For establishing Mechanism Design Theory (how institutions achieve desirable economic outcomes when information is private).
2010	Peter A. Diamond, Dale T. Mortensen, Christopher A. Pissarides	For analyzing markets with search frictions (helping explain why high unemployment can coexist with job vacancies).
2016	Oliver Hart, Bengt Holmström	For foundational contributions to Contract Theory (analyzing executive pay, performance contracts, and privatization).

Year	Laureates	Contribution Summary
2002	Daniel Kahneman, Vernon L. Smith	For integrating psychological research into economic decision-making under uncertainty (Kahneman) and establishing empirical laboratory experiments (Smith).
2017	Richard Thaler	For incorporating psychologically realistic assumptions into economic decision-making (Behavioral Economics).

Year	Laureates	Contribution Summary
2021 (Economics)	David Card, Joshua Angrist, Guido Imbens	For empirical contributions to labor economics (Card) and methodological breakthroughs in analyzing causal relationships using natural experiments (Angrist/Imbens).
2024 (Physics)	John J. Hopfield & Geoffrey E. Hinton	Foundational Neural Network Architectures: Awarded for foundational discoveries that enable machine learning with artificial neural networks. Hopfield created an associative memory (Hopfield Network) that can store and reconstruct patterns. Hinton built on this to invent the Boltzmann machine, paving the way for deep learning architectures that underpin modern generative AI models.

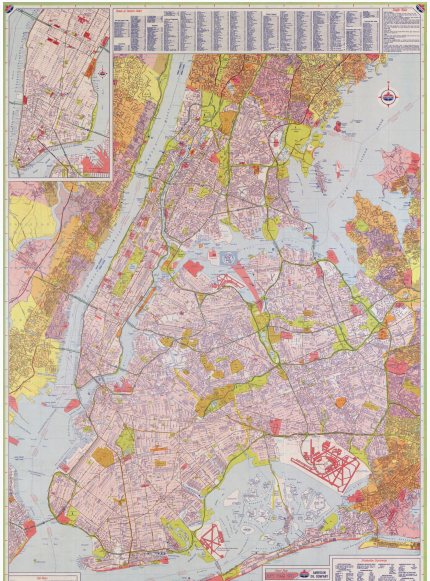
2024 (Chemistry)	Demis Hassabis & John M. Jumper (shared with David Baker)	AI for Protein Structure Prediction: Awarded for developing **AlphaFold2**, an AI system that solved a 50-year-old grand challenge in biology by predicting 3D structures for virtually all known proteins from their amino acid sequences. This revolutionized molecular biology, drug discovery, and biochemical research.

“Not only is it not right, it is not even wrong.” — Wolfgang Pauli (Peierls (1960))

Why do we build models that are clearly wrong?

- Wolfgang Pauli's comment was made in the context of a junior researcher's paper: in physics, models are approximations and hence should be falsifiable.
- Models are simplified representations of reality that are *decision-making aids or tools*.
- This is certainly true for physics, where the choice of model depends upon the problem, time and distance scales — from the Planck scales of 10^{-43} seconds (the Planck time) and lengths around 10^{-35} meters (the Planck length) to the cosmological scales where the age of the universe is 10^{10} years and the observable universe has a diameter of around 10^{11} light years.

Example — 1963 Map of New York



- This map provides a lot of information about New York City, including the location of streets, parks, and buildings.
- The subway system is indicated in small print.
- Hence it is “difficult” to use the map to efficiently navigate the subway stations.

Example — Subway Map of New York

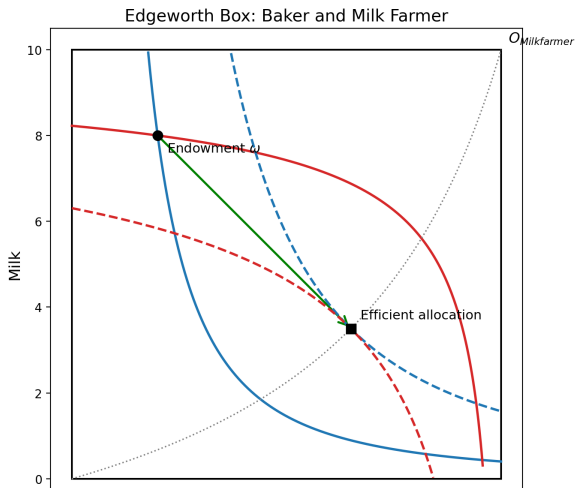


- This is a “wrong” map, but it provides a much more useful guide to choosing a subway route.
- This is a feature of *all* models - they are simplifications of the real world that allow humans to make better decisions.
- A first-order question in many contexts is how we evaluate decision quality.

Standard Economic Models

Price Theory (see appendix of MacLeod (2022))

- When we add prices and require supply = demand, then for any endowment there exists a market allocation (unique in this example).



Price theory is useful, even though it is “wrong”!

- Price theory provides a framework for understanding the allocation of commodities when they are well defined.
- It provides solutions for a number of important policy issues, including pollution (via taxes or tradable permits), congestion pricing (as in London and NYC), and the allocation of property rights via auctions (as in the FCC spectrum auctions, or timber auctions).

Recent Empirical Papers Using Price Theory

- Chetty (2011) - Empirical analysis of tax policy using price theory.
- Goolsbee (2000) - Empirical study of consumer behavior and pricing.
- Einav and Levin (2014) - Review of empirical industrial organization using price theory.

Information Economics (see chapters 2-8 of MacLeod (2022))

- Model building algorithm:
  - What are the relevant states of Nature (Savage (1972 (first published 1954)))?
  - What are the actions and payoffs available to the decision makers?
  - What are the beliefs of the decision makers over states and actions?
  - Individuals choose equilibrium strategies (e.g., Markov perfect equilibrium or sequential equilibrium).

This theory has been very successful in extending the range of economic analysis.

- Milgrom and Roberts (1992) - Amazing review of how information economics explains organizations.
- Baird and Gertner (1994) - Use information economics to explain many legal rules.
- Gennaioli and Shleifer (2007) - models the evolution of law.
- Listokin (2010) - uses Bayesian methods to set contract damages.
- Chakravarty and MacLeod (2009) - uses information economics to explain the structure of construction contracts.

Recent Empirical Papers Using Information Economics

- Einav and Levin (2008) - Empirical analysis of information asymmetries in online markets.
- Crawford and Shin (2007) - Empirical study of signaling and screening in auctions.
- Aguirregabiria (2010) - Empirical investigation of dynamic games with incomplete information.

Economics for the Age of Computers - Bounded Rationality and Computational Complexity

Bounded Rationality and Computational Complexity

- Computational complexity measures the amount of time required to achieve a particular solution.
- Many problems, such as NP-hard problems and problems with exponential complexity, can only be solved approximately within reasonable time limits.
- Bounded rationality is the term used in economics to describe decision-making under cognitive limitations.
- Standard economic models consider physical resource constraints and information constraints; models with bounded rationality add cognitive constraints.

A Cognitive Model of Bounded Rationality (Simon (1955), MacLeod (2016))

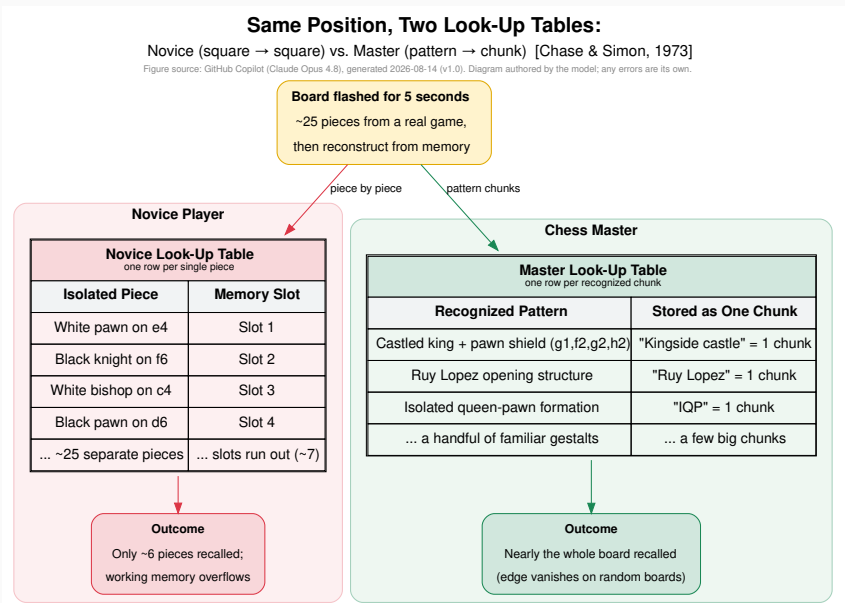
Chess Skill

- Chess is a game of perfect information, with a well-defined set of possible states and outcomes.
- Every board position defines the state of the game.
- Given the state, the player whose turn it is chooses a legal move to change the state of the game.
- The winner is the player who checkmates the other player's king.
- There are also rules that ensure that the game always ends, and in some cases a draw is possible.

Chess Playing is an Occupation - what do our previous models have to say about it?

- First, note that there is no uncertainty regarding the rules.
- We can study the game using the tools of information economics - specifically game theory:
  - Zermelo proved that in a game of perfect information like chess, either one player has a winning strategy or both have a drawing strategy.
  - Since the game is finite, and parties observe the state at each stage, the optimal play can be found by backward induction from the end.
- Yet the game tree has roughly 10^{120} paths (Shannon's estimate), so backward induction is computationally infeasible for any player, human or machine. (Note that the number of atoms in the universe is about 10^{80}.)
- Hence, given that the world is much more complex than chess, humans must clearly, in the sense of Savage, use small-world models to make decisions.

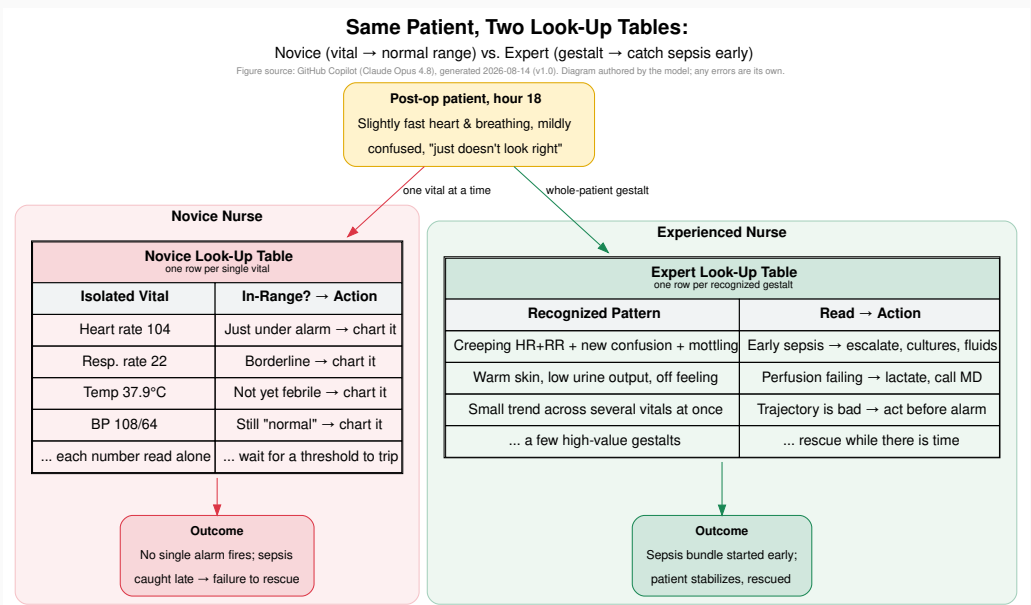
So, what do we know about chess skill? Chase and Simon (1973)



Implications

- Chess skill is a form of human capital that is acquired through experience and practice.
- The skill is not a function of the number of possible states, but rather the ability to recognize patterns and to recall the best moves for those patterns.
- This is a form of knowledge that is not easily codified or transmitted, but rather is acquired through learning and practice.
- This has implications for the theory of value: when a player faces a position that is *not* in the lookup table, the choice is guided by an evaluation function (“intuition”) acquired through experience. This *value function* is what fills the gaps in the table.

Medical Skill: The Case of Nursing



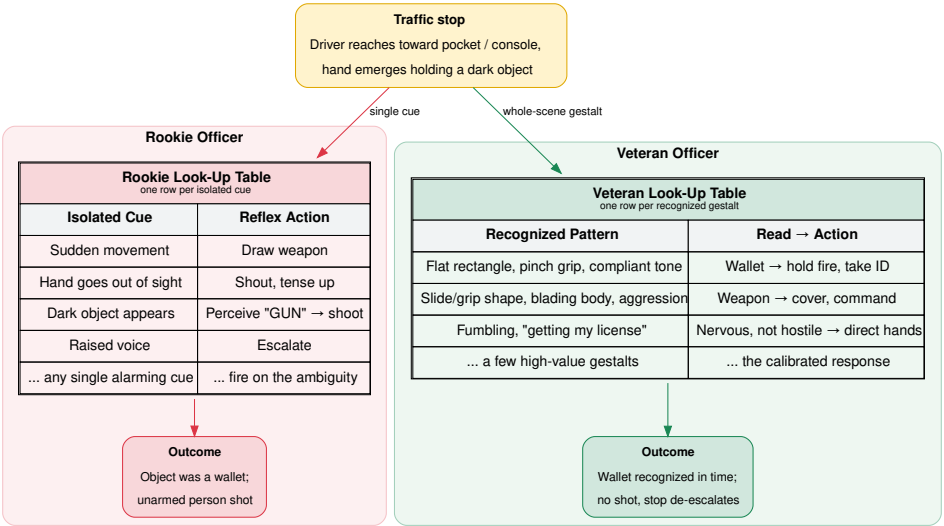
Implications

- This example illustrates that nursing skill can affect patient outcomes.
- When an adverse medical event occurs, this may lead to a lawsuit against the hospital.
- The notion of skill is relevant because tort law holds professionals to the standard of care of a reasonably skilled member of their profession (and specialists to a higher standard) when deciding whether there was negligence.

Same Stop, Two Look-Up Tables:

Rookie (cue → shoot) vs. Veteran (gestalt → recognize object)

Figure source: GitHub Copilot (Claude Opus 4.8), generated 2026-08-14 (v1.0). Diagram authored by the model; any errors are its own.



Implications

- This example illustrates that police skill can affect public safety and the outcomes of law enforcement actions.
- When an incident occurs, the skill of the responding officers may determine the resolution and any legal consequences.
- The current U.S. standard for the use of deadly force is “objective reasonableness”: whether a reasonable officer on the scene would have perceived an imminent threat (*Graham v. Connor*, 1989) - not the officer’s subjective belief.

These examples illustrate that both price theory and information economics are not sufficient.

- In all these cases, providing performance rewards at the moment of choice cannot improve outcomes: if the correct action is not in the decision maker's lookup table, no reward can elicit it.
- Rather, optimal choice depends upon rewarding *skill acquisition* ex ante.
- How should we think about aligning AI with human values ("Asilomar AI Principles" (2017))?

Empirical Papers Using Bounded Rationality i

- Peters et al. (2006) - shows that numerical skill translates into better decision-making in economic contexts.
- Banks et al. (2011) - Shows that skill affects pension savings and financial decision-making.
- van Rooij et al. (2012) - Shows that financial literacy and skill affect investment behavior and financial decision-making.
- (**?**) - Wealthy individuals are better at utility maximization.
- Choukhmane et al. (2025) - Shows that there is great variation in family decision-making skill.
- Currie and MacLeod (2020) - Shows that searching over alternatives is important for complex medical decisions.
- Griffiths et al. (2015) - Shows that patterns observed in behavioral economics can be explained as a combination of Bayesian choice and limited resources.

Bounded Rationality Implies Judicial Discretion (Hart’s Penumbra)

- The rule of law that guides modern economies relies on laws that are enforced by courts and informal arrangements.
- Similarly, modern contract theory is concerned with the *ex ante* design of reward structures that incentivize skill acquisition.
- The purpose of this section is to highlight the fact that rules are necessarily incomplete. In fact, contractual incompleteness can explain the *AI winters* that occurred from 1974 to 1980 and from 1987 to 1993.
    - Brittle Systems (Lack of Generalization): Early AI relied heavily on hand-coded rules (symbolic logic). When exposed to real-world edge cases outside their programmed logic, the systems broke down completely.

The “Open Texture” of language (Hart (1958))

- Hart (1958) introduced the idea that courts are often faced with cases that fall into the penumbra of uncertainty, where the application of legal rules is not clear-cut.
- This has a number of important implications:
  - It is impossible for judges to simply “apply the law”.
  - As with the chess player facing a novel position, judges must necessarily attach “human values” to their choices.
  - Like the police officer, judges must make a choice whether or not they are sure what is correct.

The Law as a Lookup Table

- Let $\mathcal{C}$ be the set of possible cases (fact patterns). It is effectively unbounded: new technologies, new relationships and new facts arrive continuously.
- A legal system K (statutes, precedents, doctrines) is a *finite* text that humans can in principle read, with description length $|K|$ bits.
- $Law(K, c) \in \{\mathsf{P}, \mathsf{D}, \perp\}$ is the outcome K determines for case $c \in \mathcal{C}$: plaintiff, defendant, or *undetermined* ($\perp$).
- $Just(c) \in \{\mathsf{P}, \mathsf{D}\}$ is the outcome that society's values deem correct - the target that K is trying to encode.
- In the language of the chess example, K is the *lookup table* and $Just$ is the *evaluation function*.

Proposition: Every Finite Legal System has a Penumbra

- **Proposition.** For any finite K, the *penumbra* $\Pi(K) = \{c \in \mathcal{C} : Law(K, c) = \perp \text{ or } Law(K, c) \neq Just(c)\}$ is non-empty.
- **Argument.** A text of $|K|$ bits can distinguish at most $2^{|K|}$ classes of cases. If $Just$ varies across more classes than that - as it must when $\mathcal{C}$ is unbounded - then some cases are either undetermined or misclassified. K is a *lossy compression* of $Just$.
- **Corollaries.**
  1. Hart's penumbra is not a defect of drafting; it exists for *every* finite legal system.
  2. Adding rules shrinks $\Pi(K)$ but never eliminates it, and each new rule creates new boundary cases.
  3. In $\Pi(K)$ the decision-maker must supply a value function - "human values" - to choose between P and D.
  4. An AI system's weights are also a finite K. Hence complete specification of human values is impossible (Susskind (1987) and Prakken (1997)); the residual must be handled by *discretion* and by *selecting who exercises it*.

Implications

- The proposition is an existence proof for the cases Hart (1958) places in the penumbra of uncertainty.
- For such cases the judge necessarily relies on their *human values* to make a decision.
- Hart illustrates his point with the example of a law that states “No vehicles are allowed in the park.”
- Hart points out that it is not clear how a judge would rule on bicycles, roller skates, toy automobiles, and airplanes.

AI Decisions meet Hart's Penumbra

- We conduct an experiment to see how AI systems might deal with this question.
- We provide several AI systems with the prompt:
  - "Is a {vehicle} a vehicle? On the first line, reply with exactly yes, no," "or uncertain. On the second line, provide the probability that your answer" "is correct." where vehicle is one of car, bicycle, scooter or stroller.

We vary the problem by adding the following prompts in some cases:

- JUDGE_SENTENCE = “Imagine that you are a judge.”
- RULE_SENTENCE = “There is a rule that says: no vehicles in the park.”
- For each combination of prompts, we ask four AI systems to respond yes or no and report a probability.

The AI Models Used

Model tag	Model name	Origin
`gemma3:12b`	Gemma 3 (12B)	Google (United States)
`llama3.1:8b`	Llama 3.1 (8B)	Meta (United States)
`phi4:14b`	Phi-4 (14B)	Microsoft (United States)
`qwen3:8b`	Qwen 3 (8B)	Alibaba Cloud (China)

Vehicle: car

Table 1: Mean Decision by model and judge x rule: car

Model	Judge: No / Rule: No	Judge: No / Rule: Yes	Judge: Yes / Rule: No	Judge: Yes / Rule: Yes
gemma3:12b	1.000	0.951	0.997	0.841
	(0.000)	(0.215)	(0.052)	(0.367)
llama3.1:8b	1.000	1.000	1.000	1.000
	(0.000)	(0.000)	(0.000)	(0.000)
phi4:14b	1.000	1.000	1.000	1.000
	(0.000)	(0.000)	(0.000)	(0.000)
qwen3:8b	1.000	1.000	1.000	1.000
	(0.000)	(0.000)	(0.000)	(0.000)

Vehicle: bicycle

Table 2: Mean Decision by model and judge x rule: bicycle

Model	Judge: No / Rule: No	Judge: No / Rule: Yes	Judge: Yes / Rule: No	Judge: Yes / Rule: Yes
gemma3:12b	1.000	0.000	1.000	-0.189
	(0.000)	(0.000)	(0.000)	(0.392)
llama3.1:8b	1.000	1.000	0.995	1.000
	(0.000)	(0.000)	(0.073)	(0.000)
phi4:14b	1.000	0.265	0.386	0.303
	(0.000)	(0.442)	(0.488)	(0.460)
qwen3:8b	1.000	1.000	1.000	1.000
	(0.000)	(0.000)	(0.000)	(0.000)

Vehicle: scooter

Table 3: Mean Decision by model and judge x rule: scooter

Model	Judge: No / Rule: No	Judge: No / Rule: Yes	Judge: Yes / Rule: No	Judge: Yes / Rule: Yes
gemma3:12b	1.000	0.000	1.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)
llama3.1:8b	1.000	0.981	0.951	1.000
	(0.000)	(0.136)	(0.215)	(0.000)
phi4:14b	1.000	0.243	0.784	0.211
	(0.000)	(0.430)	(0.412)	(0.408)
qwen3:8b	1.000	1.000	1.000	1.000
	(0.000)	(0.000)	(0.000)	(0.000)

Table 4: Mean Decision by model and judge x rule: stroller

Model	Judge: No / Rule: No	Judge: No / Rule: Yes	Judge: Yes / Rule: No	Judge: Yes / Rule: Yes
gemma3:12b	-1.000	-1.000	-1.000	-1.000
	(0.000)	(0.000)	(0.000)	(0.000)
llama3.1:8b	0.554	0.724	0.927	0.949
	(0.742)	(0.551)	(0.260)	(0.221)
phi4:14b	-1.000	-0.070	-0.703	0.043
	(0.000)	(0.256)	(0.458)	(0.204)
qwen3:8b	1.000	-1.000	1.000	-0.951
	(0.000)	(0.000)	(0.000)	(0.309)

How should judges resolve such cases?

- These results illustrate the following points:
  - AI systems are based on classifiers that can produce a probability for the right decision. Unlike rule-based systems, they can make a choice in any situation.
  - The evidence suggests that this comes at a cost: rule-of-law systems lose some determinacy, and the same situation may result in different outcomes depending on the AI system's interpretation and the specific circumstances.

How should we address this problem of indeterminacy?

- It has been suggested that AI systems could assist in making legal decisions, as in the case of smart contracts.
- The question here is how would we want an AI system to resolve the vehicle case?
- The “Asilomar AI Principles” (“Asilomar AI Principles” (2017)) argue that these decisions should reflect human values.
- The challenge is that there are a number of principles involved here, including the rule of law, the ordinary meaning of words, and how park users are affected by the law.

The Current Situation

- AI systems are rapidly developing, so one cannot at a given point in time make a statement that will not soon change.
- One can say that recent examples of rogue AI behavior illustrate the challenge of setting goals that result in unintended consequences.
- In each of these recent cases, the agents were given goals but then acted to achieve them in unexpected ways.

Hoping for A While Rewarding B - Kerr (1975)

- This is a well-known problem in agency theory.
- The Kerr study provides a number of examples where performance measures were designed to achieve specific goals—exactly the goal of AI alignment.
- This is a well-known problem for which there is no solution - Prendergast (1999)
- Prendergast cites MacLeod and Parent (1999), where we find empirically that performance measures only work in the simplest cases.
- In complex environments, performance is typically subjective and evaluated after the fact.

The Law's Solution

- Notice that AI firms are attempting to build the "right" incentive model. This is similar to the large literature in law and economics on how judges fill gaps in incomplete contracts (e.g., majoritarian defaults, or "what the parties would have agreed upon").
- What the law actually does is use an (imperfect) *selection* mechanism:
  - Judges are chosen based on their ability to interpret and apply the law and on their being of "good character."
  - My work with Elliott Ash (Ash and MacLeod (2015) and Ash and MacLeod (2021)) shows not only that judges have an intrinsic ability to do legal work, but also that non-partisan selection mechanisms can improve the overall quality of the judiciary.

Implications for AI Alignment

- It is increasingly clear that modern AI systems are so complex that even their designers do not really understand their behavior.
- Thus, rather than attempting to specify criteria for all AI systems, we might want to consider how best to *select* systems with desirable properties.
- The key issue is control rights - not over the design of systems, but over the building of selection systems, in much the way we do now for judges.
- The point is quite general - in practice most positions that give individuals control rights are filled via a selection process.

Concluding Remarks

Summary I

- All socio-economic models are simplifications of complex environments.
- The choice of model depends upon the question being asked.
- In this lecture I have discussed the three classes of models widely used in economics:
  - Price Theory
  - Information Economics
  - Bounded Rationality
- Economics can contribute by focusing on which type of model is best for a particular question.
- We also need more work on the quality of selection mechanisms for experts, which may in turn provide insights into how to manage AI systems effectively.

Summary II - Research Implications

- More work is needed to evaluate judicial selection mechanisms in different contexts and jurisdictions worldwide.
- One is reminded of the great jurists such as Oliver Wendell Holmes, Benjamin Cardozo and Learned Hand, who emphasized the importance of experience and practical wisdom in judicial decision-making.
- Thus, the AI alignment question might be not so much what is the correct algorithm or learning system, but how can we build selection processes that choose systems with the experience and practical wisdom that would allow us to delegate to such a system important life-and-death decisions.
- This group has both the tools and wisdom to tackle these complex issues.

Many Thanks!

- Please send comments to wbmacleod@wbmacleod.net.

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